FRAMEWORK FOR AI USE CASE INTEGRATION IN CURRICULUM DELIVERY

Part A: AI Use Case Profile

1. **Use Case ID:** KJS-SRS-01
2. **Use Case Title:** SomaiyaSat & SomaiyaPod: A PocketQube Mission featuring Autonomous AI-Based Inter-Satellite Data Routing and Advanced Multi-Mode Amateur Radio Payloads (M17, Codec2, SSTV, & TT&C / Housekeeping)
3. **AI Application Vertical:** Space Technology and Remote Sensing
4. **Collaborating Organization:** ReOrbit, Finland
5. **Beneficiaries:** Global amateur radio (HAM) community,
6. **Problem Statement:** Small satellites such as PocketQubes have very limited power, size, communication bandwidth, and onboard computing capability. These limitations make it challenging to ensure reliable communication, confirm successful deployment, and manage multiple payload operations using traditional manual control methods. In this mission, SomaiyaSat, the PocketQube satellite, is deployed into orbit using SomaiyaPod, a dedicated satellite deployer. An intelligent and autonomous system is required to verify successful deployment, manage the transition from deployer to satellite operations, and efficiently transmit telemetry, payload and amateur radio data to ground stations. The system must also support multiple communication modes, including M17 digital voice, Codec2 voice, SSTV image transmission, and telemetry and command (TT&C) data, while operating within limited communication opportunities and a restricted power budget.
7. **Background and Context:** PocketQubes are a new generation of ultra-small satellites, typically built around a 5 cm cube unit. They are widely used in educational, research, and amateur satellite missions because they are cost-effective and can be developed in a relatively short time. In this mission, SomaiyaSat is the main PocketQube satellite carrying multiple amateur radio payloads, while SomaiyaPod serves as the deployer that safely holds the satellite during launch and releases it into orbit at the required time. After deployment, confirmation signals such as separation switches, timers, or short-range communication links help verify successful release and activate SomaiyaSat's onboard systems. Once operational, SomaiyaSat supports several communication modes, including M17 digital voice and data transmission, Codec2 low-bitrate digital voice communication, SSTV image transmission, and Telemetry, Tracking & Command (TT&C) data for monitoring the satellite's health and status. Managing these different communication modes over a limited radio link, particularly during the short communication windows available in Low Earth Orbit (LEO), requires intelligent and autonomous decision-making, making it an ideal application for AI-driven mission management.
8. **Objectives:**
    1) To develop, qualify, and validate the SomaiyaPod deployer system by demonstrating safe satellite deployment, reliable deployment confirmation, and successful post-deployment spacecraft initialization in accordance with mission requirements.
    2) To design, develop, and validate a PocketQube satellite capable of autonomous mission operations through onboard AI-based decision-making for data prioritization, communication management, and resource allocation, while demonstrating reliable performance under varying communication, power, and operational conditions.
    3) To demonstrate reliable amateur radio communication services through a shared communication architecture supporting telemetry, telecommand, digital voice, and SSTV image transmission between the spacecraft and ground stations.

4) To verify the functionality, reliability, and survivability of the spacecraft through environmental qualification, system integration, communication validation, and end-to-end mission operations testing, culminating in the successful demonstration of deployment, autonomous operations, payload functionality, and ground-station communications.
5) To establish a multidisciplinary platform for student-led research and hands-on learning in artificial intelligence, embedded systems, RF and satellite communications, aerospace engineering, mechanical system design, and space mission operations through the development, integration, and operation of the mission.

9. Solution Approach:

The mission architecture separates the problem into three cooperating subsystems.

1) SomaiyaPod, the deployer, houses SomaiyaSat through launch ascent and executes a controlled release at the planned orbital insertion point, sending a deployment-confirmation signal (e.g., via separation switch or short-duration RF beacon) that SomaiyaSat uses to initialize its on-board systems and begin its commissioning sequence.
2) An onboard AI-based scheduler/router aboard SomaiyaSat (running on a low-power microcontroller or SoC) ingests link-quality estimates, battery/power telemetry, and a prioritized queue of pending data (TT&C/housekeeping always highest priority, followed by SSTV images, then Codec2/M17 voice or data traffic) and dynamically decides transmission order and mode switching to maximize successful data delivery within each ground pass.
3) The multi-mode RF payload stack implements M17 and Codec2 for digital voice/data, SSTV encoding for image downlink, and a dedicated TT&C/housekeeping channel, all multiplexed under the AI router's control. The AI models themselves are designed to be lightweight (e.g., decision trees, small neural networks, or reinforcement-learning-based policies trained on simulated link/power scenarios) so they can run reliably on resource-constrained PocketQube hardware.

10. Operational Challenges:

1) Ensuring reliable, fail-safe deployment of SomaiyaSat from SomaiyaPod, including confirmation signaling and contingency handling if deployment is delayed or partial.
2) Designing AI/ML models compact and efficient enough to run within PocketQube-class power (often sub-1W average) and compute budgets.
3) Balancing competing data priorities (real-time TT&C vs. bandwidth-heavy SSTV imagery vs. voice/digital traffic) under unpredictable link conditions.
4) Verifying autonomous routing decisions are safe and fail-gracefully (i.e., the satellite must never lose contact or fail to deliver critical housekeeping data due to an AI scheduling error).
5) Testing and validating multi-mode radio payload switching (M17/Codec2/SSTV/TT&C) without access to a live orbital environment before launch.
6) Managing thermal, power, and structural constraints common to PocketQube form factors while accommodating multiple RF payload modules, plus designing a deployer mechanism (SomaiyaPod) that reliably survives launch loads and releases SomaiyaSat without damage or tumbling.
7) Coordinating regulatory and amateur radio licensing requirements for the multi-mode payloads.

11. High-Level Workflow:

A. Mission Requirements & Interface Definition

- Define overall mission requirements for the SomaiyaSat–SomaiyaPod system.
- Specify the deployer (SomaiyaPod) and satellite (SomaiyaSat) interface (led by aerospace/mechanical and electronics teams).
- Define the mission concept, objectives, requirements, and success criteria for the SomaiyaSat–SomaiyaPod mission, including educational, research, amateur radio, and technology-demonstration goals and prepare interface control documents (ICDs).
- Perform mission analysis and system design using industry-standard tools such as STK (Systems Tool Kit) and GMAT (General Mission Analysis Tool) for orbit design, access analysis, communication link budgeting, mission planning, and operational scenario evaluation.

B. AI Model Design & Training

- Design AI models for autonomous data routing and prioritization.
- Train models on simulated link-quality, power, and data-priority scenarios (led by AI/Computer Engineering teams).
- Design and demonstrate an AI-based autonomous data routing and link-management system aboard SomaiyaSat that decides what data (TT&C, housekeeping, SSTV imagery, voice/digital payloads) to prioritize and transmit to ground stations, based on link quality, power state, and data urgency.
- Build lightweight, onboard-compatible AI/ML models capable of running within the power, memory, and compute budget of a PocketQube.

C. Payload & Deployer Development (Parallel Track)

SomaiyaSat Payload Development

- **Component Selection:** Select RF transceiver modules, antennas, microcontroller/SoC for the flight computer, power management ICs, and sensor suites (camera module for SSTV, telemetry sensors for housekeeping) based on power budget, mass/volume constraints of the PocketQube form factor, and radiation/thermal tolerance.
- **Firmware Development:** Develop firmware for each payload mode independently — M17 digital voice/data encoding-decoding, Codec2 voice compression, SSTV image capture and encoding, and TT&C/housekeeping data acquisition and formatting — followed by integration into a unified payload firmware stack that interfaces with the AI-based scheduler/router.
- **Subsystem-Level Testing:** Bench-test each radio mode independently (M17, Codec2, SSTV, TT&C) for correct encoding/decoding, signal integrity, and power draw; validate sensor readings against known references; test power management circuitry under simulated eclipse/sunlight power cycles.
- **Integration Testing:** Test all payload modules together on the SomaiyaSat bus to confirm no resource conflicts (shared RF front-end, processing time-slicing, memory allocation) and verify the AI router can correctly switch between modes under simulated link/power scenarios.
- Integrate multi-mode amateur radio payloads (M17, Codec2, SSTV, TT&C/housekeeping) onto the SomaiyaSat PocketQube platform with shared hardware/software resources.
- Conduct subsystem-level and integrated system testing, including ground-based RF communication testing, deployment and release simulations, and end-to-end operational validation.

- Establish reliable deployment confirmation and handover mechanisms between SomaiyaPod (deployer), SomaiyaSat (satellite), and the ground station to ensure successful post-deployment initialization and operations.

SomaiyaPod Deployer Development

- **Component Selection:** Select deployer structural materials, release mechanism components (e.g., burn-wire, spring-loaded pusher plate, or pyrotechnic/non-pyrotechnic actuator), separation switches, and timer/trigger electronics, chosen for compliance with PocketQube deployer standards and launch vehicle safety requirements.
- **Firmware Development:** Develop firmware/logic for the release timer or trigger sequence, separation-confirmation signaling (e.g., switch-based or short-range RF beacon), and any deployer-side housekeeping (e.g., deployer temperature, actuator status) that needs to be logged or relayed.
- **Subsystem-Level Testing:** Test the release mechanism in isolation for repeatable, jam-free actuation; verify separation switches trigger reliably; test timer/trigger firmware against expected launch-sequence inputs; conduct vibration and thermal-vacuum-representative testing on the deployer structure to simulate launch and orbital conditions.
- **Integration Testing:** Conduct full deployment rehearsals with a SomaiyaSat structural/mass simulant to confirm clean separation, correct ejection velocity/tip-off rates, and reliable receipt of the deployment-confirmation signal by SomaiyaSat's onboard systems.

D. AI Deployment & Hardware-in-the-Loop Testing

- Deploy the AI routing model onto SomaiyaSat's onboard flight computer.
- Test via hardware-in-the-loop simulation emulating post-deployment commissioning and ground-link conditions.

E. Ground Station Development

- Develop ground station software to receive, decode, and visualize multi-mode downlinked data.

F. Integrated Testing & Demonstration

- Conduct integrated system testing and environmental qualification, including deployer release testing.
- Carry out a ground/balloon/orbital demonstration to validate end-to-end autonomous operation from deployment through data downlink.
- Feed telemetry back into iterative model refinement.
- Perform Environmental Test Level (ETL) qualification testing to verify mission readiness and space survivability, including:
  i) Electromagnetic Interference (EMI)/ Electromagnetic Compatibility (EMC) test
  ii) Thermal Vacuum (Thermo-Vac) test
  iii) Vibration test

- Prepare comprehensive mission documentation and launch-readiness reports, including system design documents, assembly and integration records, test procedures, EMI/EMC, Thermal Vacuum, and Vibration test reports, RF communication test reports, deployment validation reports, safety and reliability analyses, and all technical documentation required for satellite qualification, regulatory compliance, and launch approval.

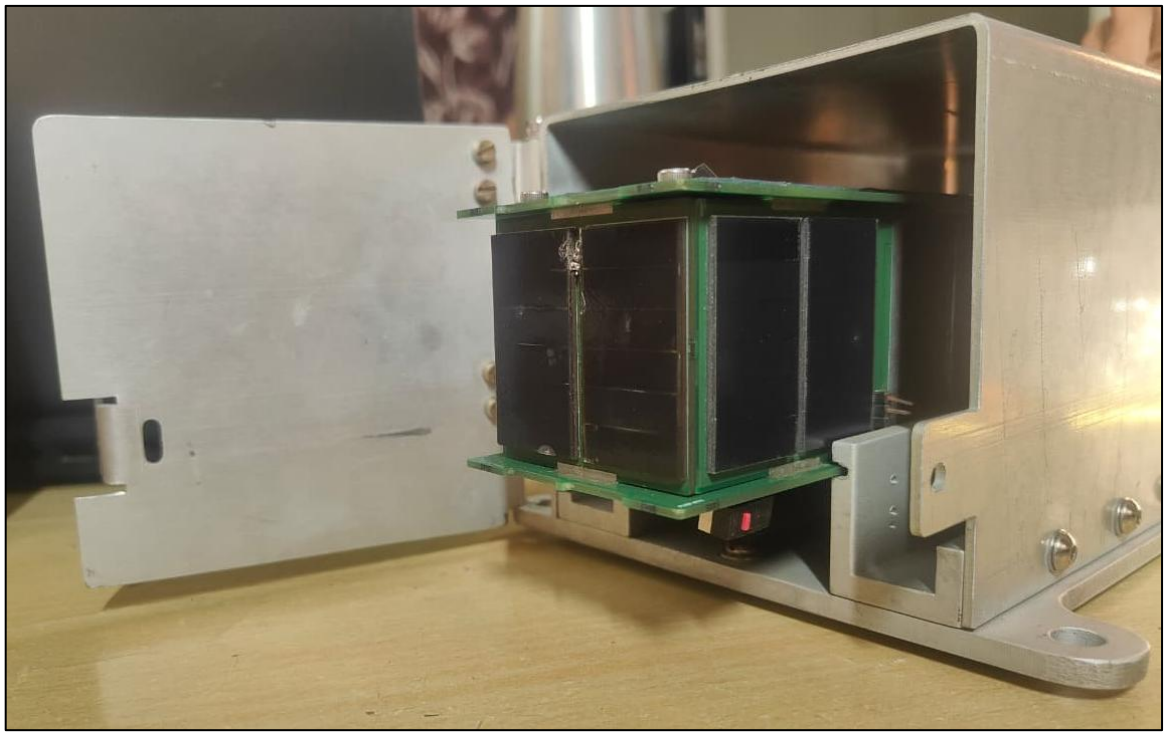


Fig. 1: SomaiyaSat and SomaiyaPOD

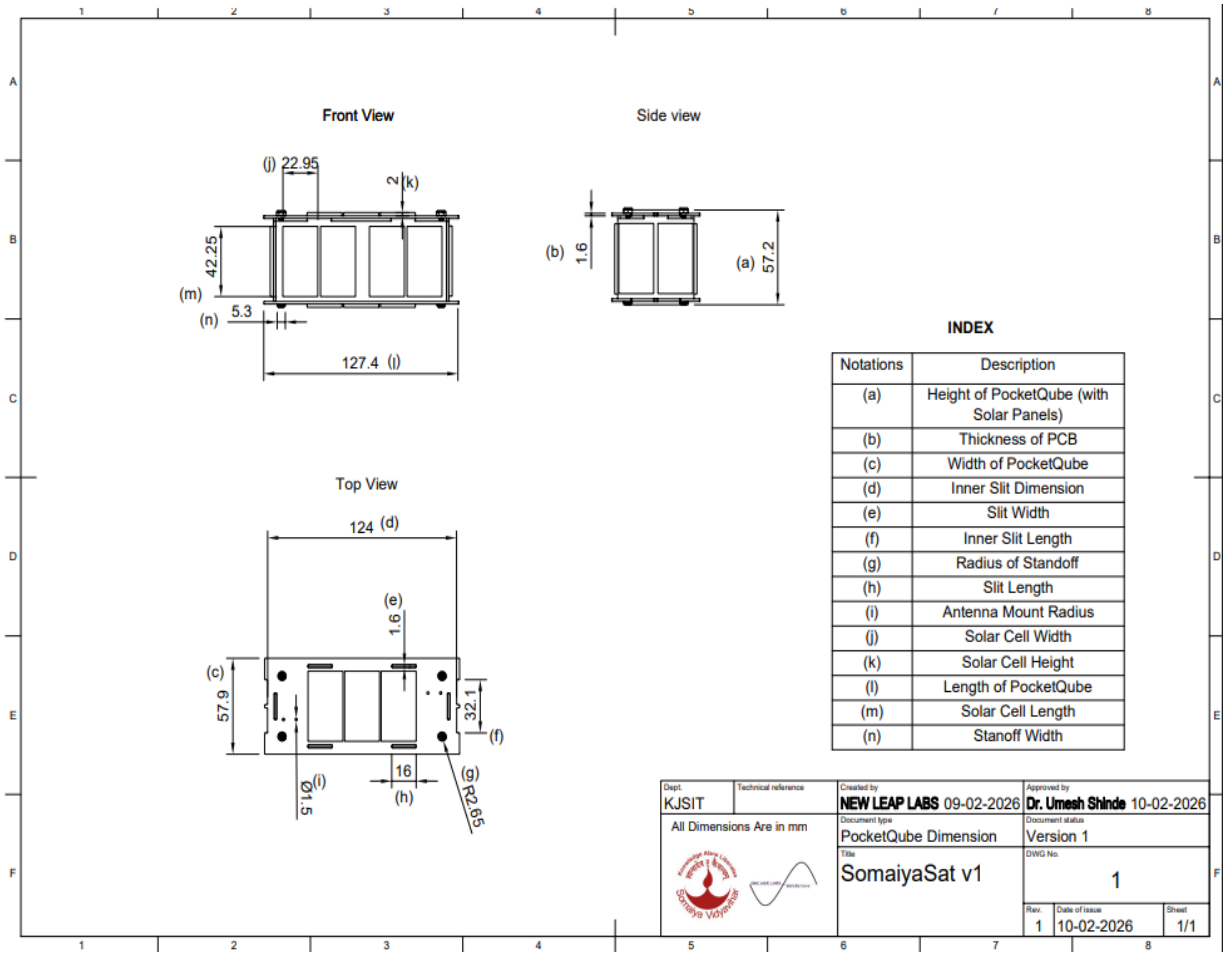


Fig. 2: Dimensions of SomaiyaSat (in mm)

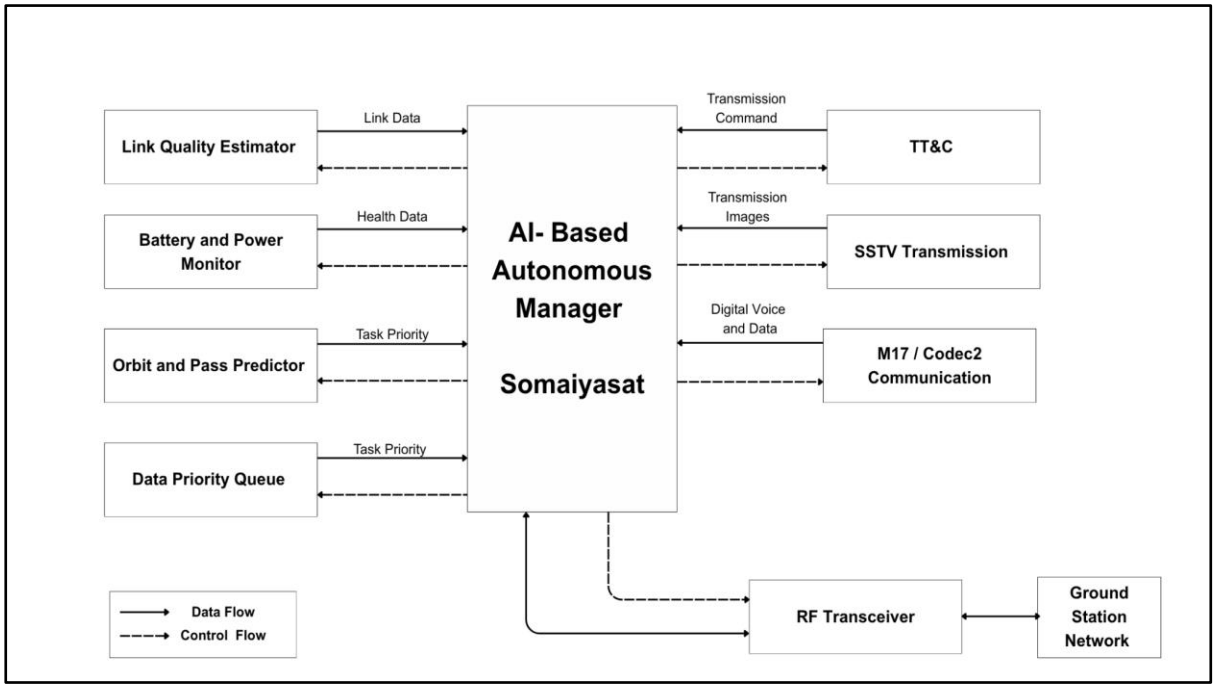


Fig. 3: Somaiyasat Architecture

12. Expected Outputs:

1) A working PocketQube-class satellite (SomaiyaSat) released by its purpose-built deployer (SomaiyaPod), demonstrating reliable deployment confirmation and AI-driven autonomous data routing post-deployment.
2) A validated multi-mode amateur radio payload supporting M17, Codec2, SSTV, and TT&C/housekeeping.
3) Onboard AI models suitable for resource-constrained space platforms.
4) Ground station software for multi-mode reception and decoding.
5) Documented performance results (deployment success rate, link success rates, data prioritization accuracy, power efficiency) usable for academic publication and as a template for future student-built satellite missions.

13. Modern AI Governance and Paradigms:

Area	Relevance	Recommended Consideration
Privacy & Data Protection	Low	Mission data (telemetry, SSTV imagery, housekeeping) is non-personal; ensure amateur radio transmissions comply with open-access norms while protecting any sensitive ground-station credentials/command links via encryption/authentication.
AI Security & Trustworthiness	High	Onboard AI routing decisions must be verifiable and fail-safe; implement watchdogs/fallback rule-based logic in case AI model output is anomalous or link conditions are unanticipated.
Responsible AI	Medium	Ensure AI decision logic is explainable to the ground team (e.g., logged rationale for prioritization choices) for debugging and trust-building.
Generative AI / LLMs	Low	Not core to onboard operation; may be used in ground-segment tools (e.g., automated report generation from telemetry logs).

Agentic AI	High	The onboard router itself acts as an autonomous agent making real-time scheduling/routing decisions without human-in-the-loop during each pass.
Human-in-the-Loop Decision Making	Medium	Ground operators retain command override capability; AI autonomy is bounded by uplinked commands and safe-mode triggers.
Regulatory / Governance Considerations	High	Amateur radio frequency allocation and licensing (per ITU/national HAM regulations), and PocketQube launch/registration compliance, must be addressed early.

Part B: Program-wise AI Use Case Mapping

14. Programs Mapped:

Institute: KJSIT		**Institute: KJSSE**	
Information Technology	√	Information Technology	√
Computer Engineering	√	Computer Engineering	√
Artificial Intelligence & Data Science	√	Artificial Intelligence & Data Science	√
Electronics and Telecommunication	√	Electronics & Telecommunication Engineering	√
		Computer & Communication Engineering	√
		Electronics & Computer Engineering	√
		Electronics Engineering (VLSI Design & Technology)	×
		Computer Science & Business Systems	√
		Mechanical Engineering	√
		Robotics & Artificial Intelligence	√

15. Technology Domains Mapped:

Program	**Technology Domain**	**Layer**	**Role in Use Case**	**Inter-Disciplinary Dependencies**
CE, IT, AI&DS	AI/ML Model Development	Software	Design and training of onboard data-routing and prioritization models	Receives link-quality/power telemetry from EXTC team; provides decisions to embedded firmware team
EXTC, ECE	RF Communications & Payload Design	Hardware/ Firmware	Implementation of M17, Codec2, SSTV, and TT&C radio modules	Supplies signal/link data to AI router; depends on power budget from Mechanical
MECH, RAI	Structural & Thermal Design	Hardware	PocketQube chassis, thermal management, mechanical integration	Constrains power and volume available to electronics and AI compute hardware
CSE	Embedded Systems & Flight Software	Software/ Firmware	Deployment of AI models on flight computer; system integration	Integrates AI/EXTC outputs into a unified onboard control loop

Part C: Course-level AI Use Case Mapping

16. Course-level AI Use Case Mapping:

Institute: Name	Program: Name		
Course	**Contribution to Use Case**	**Relevant Topics from Syllabus**	**Potential Modes of Integration**

Institute: Name	Program: Name		
Course	**Contribution to Use Case**	**Relevant Topics from Syllabus**	**Potential Modes of Integration**

Part D: Inter-disciplinary Faculty Committee

17. Faculty Owner(s):

- Dr. Umesh Shinde, Associate Professor, Department of Basic Science and Humanities, K J Somaiya Institute of Technology
- Dr. Shailesh Nikam, Professor, Department of Mechanical Engineering, K J Somaiya School of Engineering

18. Program-level Integration Coordinators:

Institute	Program	Coordinator
KJSIT	e.g. IT	Faculty Name & Designation
KJSSE	e.g. Mechanical	Faculty Name & Designation

19. Course-wise Integration Coordinators:

Institute	Program	Course	Course In-charge
KJSIT	e.g. IT		Faculty Name & Designation
KJSSE	e.g. Mechanical		Faculty Name & Designation

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